



Artificial Intelligence
For Georgia

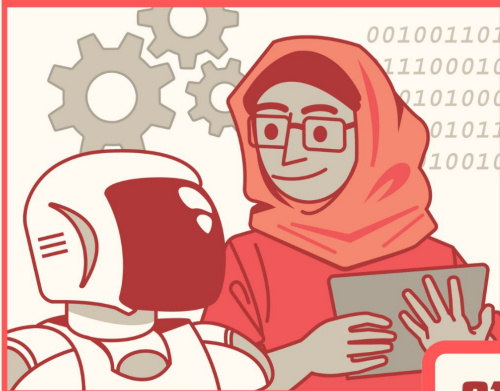
AI Careers



Developed for the AI4GA (Artificial Intelligence for Georgia) project funded by National Science Foundation awards DRL-2049029 and DRL-2048502. **This work is licensed under the Creative Commons.**

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ROBOTICS ENGINEER



01

Design and development
of robotic systems

Examples

Delivery robots
Warehouse robots
Self-driving cars

Degree Pathways

Mechanical Engineering
Electrical Engineering
Computer Engineering
Computer Science or Robotics

Technical Skills

Mechanical Design
Circuit Design
Robotics Algorithms
Programming

Median Salary

\$85,000 - \$120,000

AI ARCHITECT



08

Plans and designs large
AI-powered software
systems

Examples

Design an AI system that helps a company better understand and serve its customers

Degree Pathways

Computer Science
Information Systems & Technology
Data Science
Machine Learning

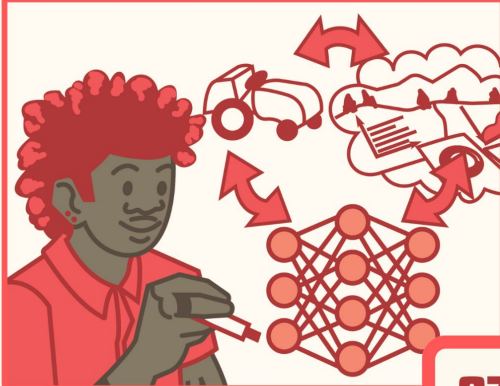
Technical Skills

System Architecture	Programming
Data Management	Machine Learning
Project Management	Cloud Computing

Median Salary

\$107,000 - \$130,000

AI RESEARCHER



07

Develops new algorithms and architectures for artificial intelligence

Examples

Developing new machine learning algorithms to train large language models such as ChatGPT

Degree Pathways

Computer Science
Machine Learning
Mathematics
Statistics

Technical Skills

Machine Learning	Neural Networks
Model Evaluation	Statistics
Algorithm Design	Programming

Median Salary

\$93,000 - \$130,000

COMPUTER VISION ENGINEER



10

Develops algorithms and applications that allow computers to recognize information in images or video

Examples

Face recognition in smart phones

Visual search (e.g., Google Lens, Pinterest)

Scene understanding in self-driving cars

Degree Pathways

Computer Science

Computer Engineering

Robotics

Technical Skills

Computer Vision

Image Processing

Machine Learning

Artificial Intelligence

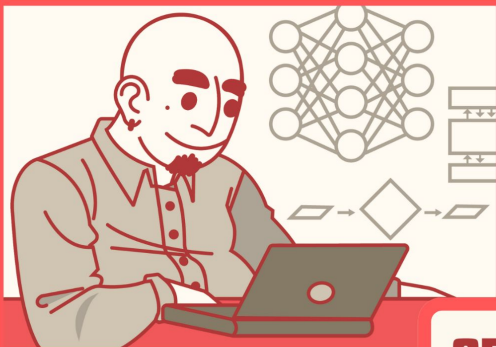
Programming

Computer Graphics

Median Salary

\$165,000 - \$170,000

MACHINE LEARNING ENGINEER



05

Designs and builds AI systems
that use large datasets to
make predictions

Examples:

Movie, music, or product
recommendation systems
Training generative AI models (e.g.,
ChatGPT, Midjourney)

Degree Pathways

Computer Science
Machine Learning
Mathematics
Statistics

Technical Skills

Machine Learning Algorithms
Deep Neural Networks
Model Evaluation
Statistics
Software Engineering

Median Salary:

\$145,000 - \$160,000

BUSINESS ANALYST



03

Solves business problems
using statistical and
machine learning tools

Examples

Performing financial analyses to
predict revenue and expenses
Forecasting seasonal demand for
products

Degree Pathways

Business Administration
Finance
Data Science
Computer Science

Technical Skills

Statistical Analysis
Financial Forecasting
Data Visualization
Machine Learning

Median Salary

\$68,000 - \$120,000

DATA ARCHITECT



09

Designs and implements an organization's strategies for data storage and access

Examples

Designing an Internet retailer's product and customer databases
Implementation of storage and security policies for public records databases

Degree Pathways

Information Systems & Technology
Computer Science
Data Science

Technical Skills

Software Development	System Architecture
Machine Learning	Data Visualization
Databases	Cybersecurity
Cloud Computing	

Median Salary

\$143,000 - \$150,000

DATA ENGINEER



02

Creates and manages large datasets to support machine learning and data science projects

Examples

Collecting and organizing disease outbreak data from multiple health agencies

Degree Pathways

Information Systems & Technology
Computer Science

Technical Skills

Data Management
Data Science
Machine Learning
Programming

Median Salary

\$151,000 - \$160,000

DATA SCIENTIST



04

Collects and analyzes datasets to detect patterns that help organizations make better decisions

Examples

Discovering new types of customers based on purchase history
Modeling changes in animal populations due to climate change

Degree Pathways

Computer Science
Statistics
Data Science

Technical Skills

Data Management
Data Visualization
Data Analysis
Machine Learning

Median Salary

\$115,000 - \$130,000

PRODUCT MANAGER



06

Manages the planning, development, and launch of products and services that use AI

Examples

Works with a team to introduce new AI features into email, social media, or graphic design tools

Degree Pathways

Business Administration
Marketing
Computer Science

Technical Skills

Business Strategy	Design Thinking
Market Research	Problem-solving
Data Analytics	Team Management
Behavior Sciences	

Median Salary

\$131,000 - \$150,000

USER EXPERIENCE (UX) DESIGNER



11

Works with users and software developers to create products that are easy to understand and use

Examples

Design the passenger experience in self-driving cars (e.g., in-car displays, audio notifications, and mobile app)

Design interfaces for mobile apps, websites, and digital products

Degree Pathways

Computer Science

Web and Graphic Design

Human Factors Engineering

Psychology

Technical Skills

Interviewing

Interface Design

Field Research

Prototyping

Data Analysis

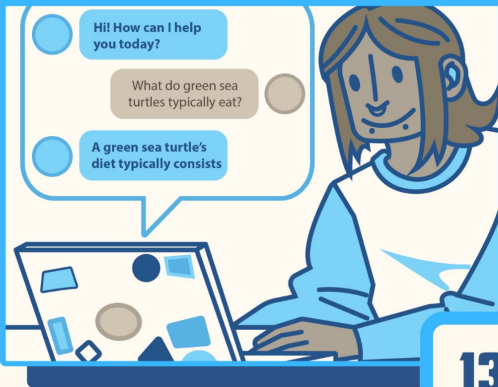
User Testing

Behavioral Analysis

Median Salary

\$143,000 - \$150,000

CONVERSATIONAL AI ENGINEER



Develops Natural
Language Processing
functionality for chatbots
and intelligent assistants

Examples

Develop customer service
chatbots for websites
Develop intelligent assistants such
as Siri, Alexa, or Google Home

Degree Pathways

Computer Science
Artificial Intelligence
Cognitive Science
Linguistics

Technical Skills

Linguistic Analysis
Dialogue Management
Large Language Models
Software Engineering

Median Salary

\$63,000 - \$85,000

NATURAL LANGUAGE PROCESSING (NLP) ENGINEER



12

Develops applications that can understand and generate human language

Examples

Develops algorithms that allow search engines to understand & answer questions
Develops text summarization and language translation apps

Degree Pathways

Computer Science
Machine Learning
Cognitive Science
Linguistics

Technical Skills

Linguistic Analysis
Machine Learning
Large Language Models
Software Engineering

Median Salary

\$108,000 - \$120,000



Artificial Intelligence
For Georgia

Applications of AI



AI in AGRICULTURE



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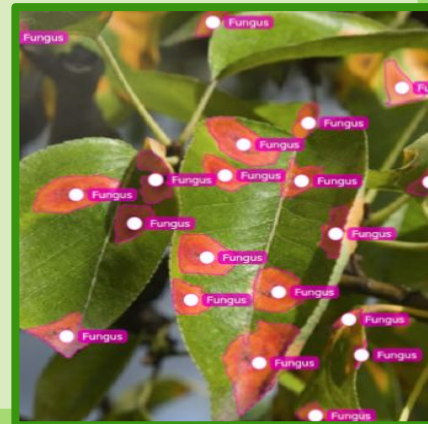


Livestock Health Monitoring



Monitoring fruit ripening

Automated crop harvesting



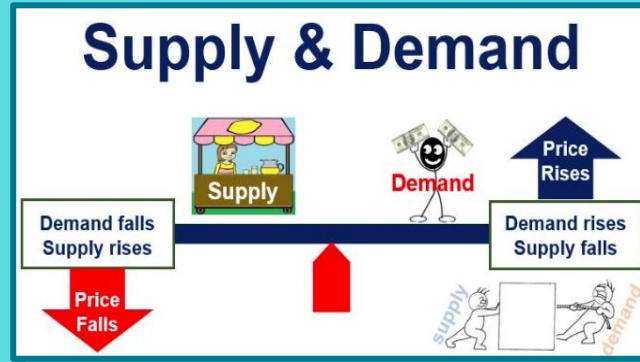
Disease Detection

AI in MANUFACTURING

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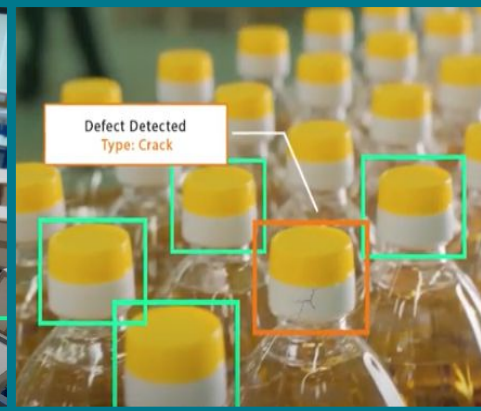
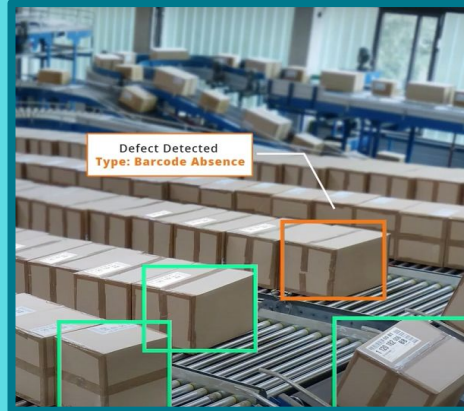
Improve Demand Forecasting



Automated defect detection



Automated Assembly Lines



AI in MEDICINE

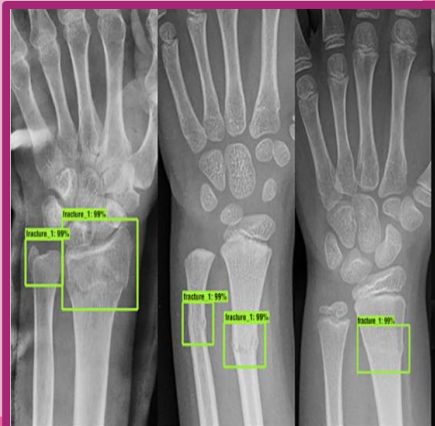
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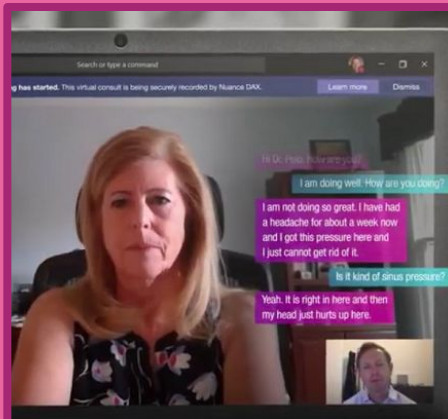
Drug Discovery



Medical Record Transcription



Automated Diagnosis



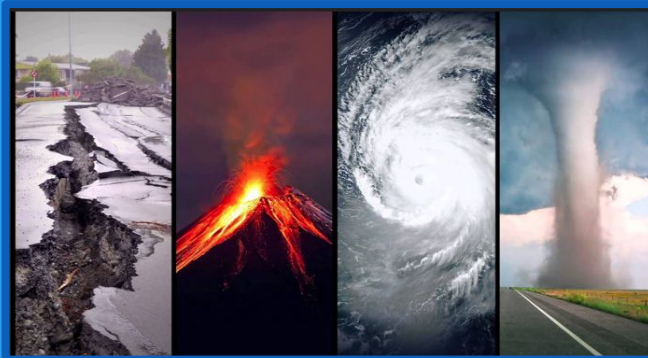
Hospital Robots

AI in PUBLIC SAFETY

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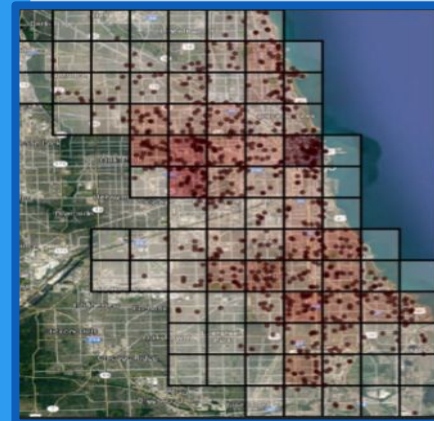
Natural Disaster Prediction



Critical Infrastructure Protection



Video/Image Analysis



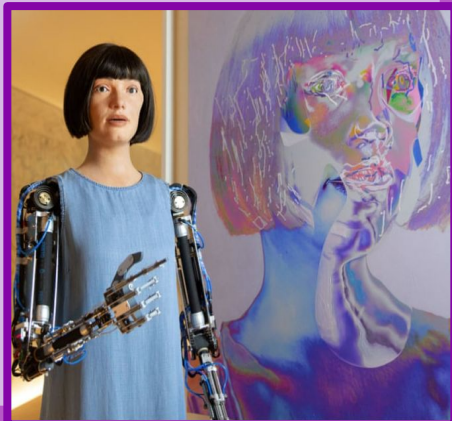
Crime Prediction

AI in ARTS

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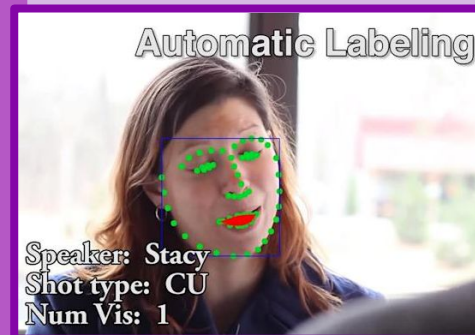


Music Composing



Artwork Creation

Virtual Characters in Movies



Scene Editing Assistance

AI in EDUCATION



Automated tutoring

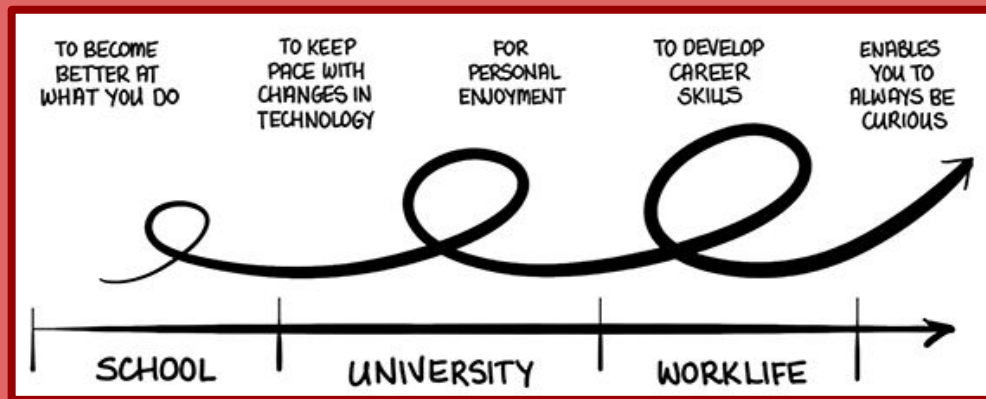


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Personalized Education
Programs

Lifelong learning and re-skilling

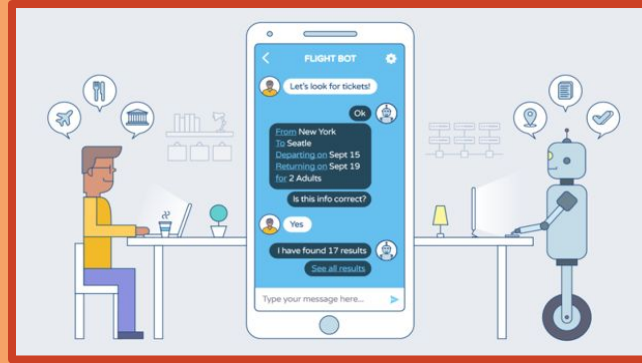


AI in MARKETING

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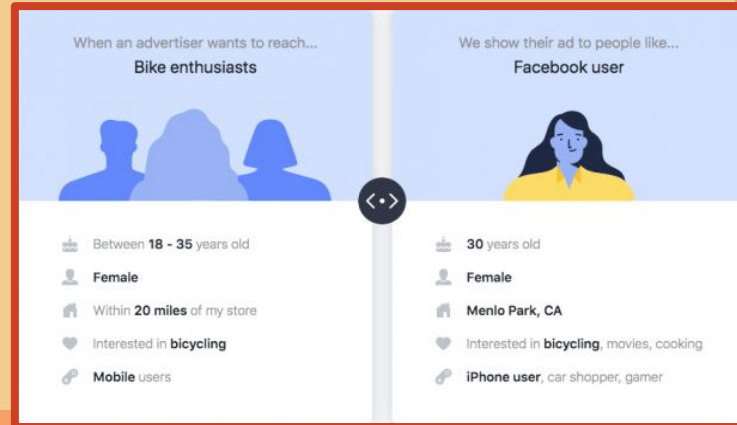
Chatbot Sales Assistants



Targeted Marketing



Consumer Needs Prediction



AI in TRANSPORTATION



Real-Time Traffic Monitoring



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Forecasting Vehicle
Maintenance

Self-Driving Cars

